AWS S3 [Simple storage service]

→ It's basically a storage where we can store data

→ Not a server so don't have any CPU, OS etc. Just storage.

→ Used to store user data, website hosting, app hosting.

→ Then मानो Google drive से सारा पढ़े? Yeah, it's cheap but personal use सारा नहीं

For development, we need faster response of data too, along with Instance [virtual server]

so, for that purpose S3 bucket is used.

S3 security is entirely dependent on IAM Policy & Roles given to user & service respectively

→ Basically a bucket in which data are stored in form of objects.

→ Reliability:- 99.99999999999 %
11 9's

- Data is replicated across multiple facilities
- Ultimate storage
- Bucket name has to be unique.

S3 Storage classes

S3 Standard	S3 IA -Infrequent Access	S3 One zone -IA	S3 Intelligent Tiering	S3 Glacier	S3 Glacier Deep Archive
-General purpose -Fast access -Replicated in multiple availability zone -costly than others	-Less Frequent accessed -~~Slow~~ Fast access -In multiple AZ	-Less accessed -slow -In one zone only	-Automatically moves data to more cost effective tier	-Low cost -For data archiving	-Lowest cost storage -slower to retrieve (12 hrs)

Does this!

To reduce cost

⇒ Litecycle Policies:-

S3 Standard $\xrightarrow{\text{After 30 days}}$ S3 IA $\xrightarrow{\text{30 days}}$ S3 One zone IA $\xrightarrow{\text{90 days}}$ S3 Glacier

↳ Can be done automatically!

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→ Enable Delete Marker, now ~~onf~~ IF (one version / Not entire object) one file is deleted from b1, same file will be deleted from b2. [replicated one] [not deleted = hidden & hidden]

→ If ~~that~~ Disable, Destination bucket don't ~~delete~~ the delete the object deleted in the source bucket.

→ Now If we delete entire (All version) object in source it don't ~~delete~~ delete the object in destination
- Keeping it as backup.

S3 Storage Types

1.] Standard / Frequent - Rapid Access
- Multiple AZs
- The most cost
- Read/write happens regularly

2.] Standard IA - Rapid Access
- Multiple AZS
- Read/write Access happens rarely.
/ Rarely accessed

3.] One zone IA - "
- One AZS
- "

Glacier Classes:

1.] S3 Glacier Instant Retrieval

- For archive data / Rarely accessed.
- Retrieval Time - millisecond
- Cost more than other glacier classes

2] S3 Glacier Flexible Retrieval

- Long-term backup with occassional access
- RT - minutes to hours
- Cheaper than above.

3.) S3 Glacier Deep Archive

- Data that is almost never accessed
- RT - 12 to 48 hours [but stored efficiently]
- Cheapest of all.

↳ How → Use Magnetic Tape
→ Write once, read rarely
→ Once written, don't consume electricity
→ Vastly cheaper than SSDs & HDDs.

S3 Intelligent Tiering:

- For unpredictable access pattern
- Shifts between different S3 classes to be cost optimized → shifts based on how frequently data is accessed.
- Done automatically.

S3 Intelligent Tiering VS Lifecycle policy

- Starts at Frequent access Tier	- Can be whatever we want / generally S3 standard at start
- Not accessed moved to cheaper classes	- "
- For unpredictable Access pattern	- For predicatable access pattern
Setup - Automatic	- Manual set
cost - Dynamic, based on access	- Might be under/over optimized
	- Lifecycle policy can be applied to specified directory of S3 bucket too! only...

Amazon S3 - Moving between storage classes

Standard
↓
Standard IA
↓
One Intelligent Tiering
↓
One-zone IA
↓
Glacier Instant retrieval
↓
Glacier Flexible retrieval
↓
Glacier Deep Archive